

Bowel Ischemia Score Predicts Early Operation in Patients With Adhesive Small Bowel Obstruction.

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The American surgeon (2022) • PMID 33502222 • PMC9057653 • DOI 10.1177/0003134820988820

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Background:

Nonoperative management of adhesive small bowel obstruction (SBO) is successful in up to 80% of patients. Current recommendations advocate for computed tomography (CT) scan in all patients with SBO to supplement surgical decision-making. The hypothesis of this study was that cumulative findings on CT would predict the need for operative intervention in the setting of SBO.

Methods:

This is an analysis of a retrospectively and prospectively collected adhesive SBO database over a 6-year period. A Bowel Ischemia Score (BIS) was developed based on the Eastern Association for the Surgery of Trauma guidelines of CT findings suggestive of bowel ischemia. One point was assigned for each of the six variables. Early operation was defined as surgery within 6 hours of CT scan.

Results:

Of the 275 patients in the database, 249 (90.5%) underwent CT scan. The operative rate was 28.3% with a median time from CT to operation of 21 hours (Interquartile range 5.2–59.2 hours). Most patients (166/217, 76.4%) with a BIS of 0 or 1 were successfully managed nonoperatively, whereas the majority of those with a BIS of 3 required operative intervention (5/6, 83.3%). The discrimination (area under the receiver operating characteristic curve) of BIS for early surgery, any operative intervention, and small bowel resection were 0.83, 0.72, and 0.61, respectively.

Conclusion:

The cumulative signs of bowel ischemia on CT scan represented by BIS, rather than the presence or absence of any one finding, correlate with the need for early operative intervention.

Introduction

Small bowel obstruction (SBO) is a common clinical entity, responsible for 12–16% of acute surgical admissions.^{1–5} Peritoneal adhesions account for the majority (50–80%) of all bowel obstructions.⁶ While the majority of adhesive SBOs (>80%) are managed without surgery, there are nearly 300 000 operations performed for SBO adhesiolysis annually in the United States.⁷ There is ongoing debate on the clinical criteria to predict failure of nonoperative management and the timing of surgical intervention. Strategies for the management of SBO have changed significantly. The adage “never let the sun set or rise on a bowel obstruction” has been replaced with individualized care of the patient with obstructive symptoms. In patients without peritonitis or signs of bowel ischemia, the use of a water-soluble contrast challenge is recommended by the Eastern Association for Surgery of Trauma (EAST). However, bowel ischemia requires timely identification and emergent operation. Clinical indicators, including fever, leukocytosis, tachycardia, pain, metabolic acidosis, and peritonitis, are inadequate to identify bowel ischemia. These

indicators only identify bowel ischemia in 40%–50% of cases, representing a significant deficiency in the current scientific literature.^{8–10} There is an urgent need to identify patients requiring operation based on initial CT scan findings without awaiting the results of the water-soluble contrast challenge.

In 2012, the EAST developed guidelines for the evaluation and management of SBO.¹¹ Current recommendations advocate CT scan in all patients with SBO to supplement surgical decision-making. However, it is unknown how CT findings change clinical management in those patients with benign abdominal exams. In previous studies, CT findings did not significantly alter management decisions in patients admitted with adhesive SBO who were initially managed with nonoperative care.^{12,13}

The hypothesis of this study was that cumulative findings on CT would predict the need for operative intervention in patients without immediate clinical findings (such as peritonitis) concerning for bowel ischemia. We sought to develop a novel Bowel Ischemia Score (BIS) that accurately predicts the need for early operative intervention in patients with SBO without clinical indications for emergent surgery.

Methods

Data Collection

This was an analysis of prospective cohort with a historical control using a SBO database over a 3-year period at Froedtert Memorial Lutheran Hospital, an academic teaching hospital in Milwaukee, Wisconsin. The prospective cohort and historical control were combined for statistical analysis. All patients admitted to or consulted by the emergency general surgery (EGS) service were evaluated. Small bowel obstruction was diagnosed by clinical findings, plain abdominal radiographic findings, and CTscan when available. A SBO protocol was initiated at our institution in January 2015 with a water-soluble contrast challenge and therefore two cohorts of patients were collected. The historical cohort included patients admitted from January 1, 2012 until December 31, 2014, managed without a SBO protocol. From January 1, 2015 to June 30, 2015 data were not collected while the new adhesive SBO protocol was being introduced based on the EAST practice management guidelines (Supplement 1).¹¹ The prospective cohort included patients from July 1, 2015 until June 30, 2017 who were managed according to the SBO protocol. This study was reviewed and approved by the Medical College of Wisconsin/Froedtert Hospital Institutional Review Board (PRO00025247).

Participants

We included patients who were over the age of 18 years and admitted to the EGS service or received an inpatient consultation by EGS. Exclusion criteria included patients with peritoneal signs, hemodynamic instability, perforation on CT scan, or nonadhesive cause of obstruction (eg, suspected cancer and patients with no prior abdominal operation) based on chart review by RM and KB.

Data Collection

The following data were collected: patient age, gender, admitting diagnosis, discharge diagnosis, comorbid conditions (current oral steroid use, active cancer, diabetes mellitus, hypertension, cardiac disease, congestive heart failure, current smoking, chronic obstructive pulmonary disease, end-stage renal disease, acute renal failure, and obesity), date and time of admission, date and time of discharge, time from CT scan to surgery, attending radiologist's reading of the CT scan, time from admission to surgery, need for surgical intervention during index hospital admission, need for small bowel resection in the operating room, in-hospital complications, and re-admission within 30 days. Additional retrospective chart review was performed by to collect supplemental variables (vital signs, laboratory values, abdominal tenderness, emesis, prior SBO admission, and duration of obstipation). Operative and nonoperative clinical decisions were made based on an established SBO clinical management protocol during the prospective portion of the study (Supplement 1).

Outcomes

The primary outcome of interest assessed was need for early surgery (within 6 hours of CT scan). The secondary outcome was need for any operative intervention during the index hospitalization.

Statistical Methods

The primary independent variable was the newly derived BIS. Candidate variables were identified based on the EAST guidelines of CT findings suggestive of bowel ischemia. Because obstipation greater than 24 hours, leukocytosis $\geq 12 \times 10^9/L$, and prior surgery for SBO have been associated with need for operative intervention,¹⁴ we performed logistic regression to determine their predictive value.¹⁵ One point was assigned to each of the six variables of the EAST guidelines of CT findings of bowel ischemia: (1) reduced bowel wall enhancement, (2) wall thickening, (3) mesenteric venous congestion, (4) mesenteric fluid, (5) unusual course of the mesenteric vasculature (swirl sign), and (6) ascites. The points are then simply added together. These criteria were confirmed based on the attending radiologist's reading of the CT scan. Calibration (Brier score) and discrimination (area under the receiver operating characteristic curve [AuROC]) were then evaluated in the model.^{16,17} The calibration was also tested with the Hosmer-Lemeshow goodness-of-fit test, which compares observed with predicted values by decile of predicted probability.

The individual variables in the BIS and operative outcomes were assessed using logistic regression. Based on the degree of univariate association between individual BIS variables and operative outcomes, the AuROC analysis was repeated using a modified, weighted BIS. Specifically, BIS was modified so that reduced bowel wall enhancement and swirl sign were assigned 2 points each. The four remaining variables were assigned 1 point each.

Statistical significance was defined as $P < .05$; all tests were two-tailed. Data were expressed as the mean $\pm$ standard deviation (SD) for normally distributed continuous descriptive variables with a normal distribution, median $\pm$ interquartile range (IQR) for continuous descriptive variables with a skewed distribution, and proportions for categorical variables. Student's t-test was performed to explore differences in the two groups for continuous variables with a normal distribution and the Wilcoxon rank sum test for continuous variables with a skewed distribution. The chi-square test was used to identify differences in outcomes for categorical variables. All data analysis were performed in Stata-MP version 15.1 (StataCorp, College Station, Texas).

Results

A total of 275 patients were identified in the database. Of those, 249 (90.5%) underwent CT scan. The overall operative rate was 28.3% with a median time from CT to operation of 21 hours (IQR 5.2, 59.2 hours). Patients in the historical cohort were significantly more likely to undergo any surgical intervention but had similar rates of early surgical intervention (Table 1).

With regard to CT findings, reduced bowel enhancement (OR 13.0; 95% CI, 3.66, 47.5) and swirl sign (OR 9.1; 95% CI, 2.72, 25.0) were significantly associated with early surgery (Table 2). Twenty patients had CT readings consistent with closed-loop obstruction. Seven of the twenty (35%) underwent early surgical intervention. This finding was not incorporated into the BIS because findings of ischemia in the setting of closed-loop obstruction are the same as other causes of ischemia. However, BIS was associated with closed-loop obstruction (OR 4.0, 95% CI 2.3, 7.0). The AuROC for early surgery was .83 (95% CI 1.9, 5.7) and .72 for any operative intervention (95% CI 1.9, 4.1). The AuROCs were not substantially changed when the analysis was repeated using the modified, weighted BIS. Therefore, the non-weighted BIS was used for simplicity of calculation.

Based on univariate logistic regression models, clinical variables (age, leukocytosis $\geq 12 \times 10^9/L$, duration of obstipation ≥ 24 hours, and prior surgery for SBO) were not significant predictors of early operative intervention (Table 3). Obstipation greater than 24 hours was the most predictive of the clinical variables and was associated with need for any surgical intervention ($P < .05$). The addition of these clinical variables did not result in significant improvement in the AuROC curves and therefore were not included in the score.

Bowel Ischemia Score 0 (n = 128, 46.5%) was the most common score in the cohort, followed by BIS 1 (n = 89, 32.3%), BIS 2 (n = 25, 9.1%), and BIS 3 (n = 6, 2.2%) (Table 4). Bowel Ischemia Score ≥ 1 was present in 120 (43.6%) patients. Most patients (166/217, 76.5%) with a BIS of 0 or 1 were successfully managed nonoperatively. Less than half (10/25, 40.0%) of patients with a BIS of 2 were successfully managed nonoperatively, and the majority of those with a BIS of 3 required operative intervention (5/6, 83.3%). The sensitivity, specificity, positive predictive value, and negative predictive value for BIS of ≥ 2 and early operation were 44.4%, 89.6%, 25.8%, and 95.4%, respectively. Bowel Ischemia Scores of 1, 2, and 3 were significant predictors of early operation (Table 4). Bowel Ischemia Score accurately predicts early surgery (Hosmer-Lemeshow P-value .28; Brier .06; AuROC .83), moderately predicts any surgical intervention (Hosmer-Lemeshow P-value .94; Brier .17; AuROC .72), and poorly predicts small bowel resection (Hosmer-Lemeshow P-value .91; Brier .20; AuROC .61) (Figure 1).

Discussion

Bowel Ischemia Score is a novel tool to predict the need for early operative intervention in the setting of SBO. Patients in the historical cohort were significantly more likely to undergo any surgical intervention but had similar rates of early surgical intervention. This was likely due to the addition of the water-soluble contrast protocol. The six CT variables included in BIS were each correlated with early operative intervention, but the composite BIS was more informative than any single CT finding. The AuROCs .72 and .83 indicate favorable discriminative ability for any operation and early surgery, respectively. A reasonable cutoff for potential early intervention or increased surveillance seemed to be a BIS of 2, although future studies will be required to validate our novel score. Bowel Ischemia Score can be used for communication and prognostication in the patient presenting with SBO.

Diagnosing strangulated obstruction is critical in the early management of SBO as it occurs in 16% of patients and mortality approaches 30%.^{18,19} Multiple clinical criteria have been proposed to identify strangulation including age, obstipation, vomiting, elevated white blood cells (WBC), fever, tachycardia, distension, abdominal tenderness, and acidosis.^{13,14,18–20} Fever, tachycardia, elevated WBC, and vomiting were not found to be useful.^{13,18–20} Age, obstipation, prior SBO, and abdominal tenderness were useful in several studies, but only obstipation was prospectively validated.²⁰ No clinical findings reached statistical significance in our study. The time point of early surgery (less than 6 hours after CT scan) may explain this difference. Clinical criteria alone are of limited value in diagnosing bowel ischemia.

The EAST management guidelines in 2012 recommend the use of CT scan (Level 1 evidence) and that patients with evidence of ischemia on imaging should undergo exploration.¹¹ Multiple CT findings are known to be associated with strangulation: reduced bowel wall enhancement, free fluid, abnormal course of the mesenteric vasculature, mesenteric edema, bowel wall thickness, diffuse mesenteric haziness, and closed-loop mechanism.^{21,22} Computed tomography features that are frequently associated with operative management and have strong concordance between radiologists include complete bowel obstruction, small bowel dilation greater than 4 cm, and transition point. However, transition point was the only significant factor predictive of operative management for SBO on multivariable logistic regression analysis. Zielinski et al found that mesenteric edema, small bowel feces sign, and obstipation were predictive of operative exploration.¹⁴ Similar to our findings, reduced bowel wall enhancement has been shown to have the highest specificity for strangulation and absence of mesenteric fluid is helpful to rule out strangulation.²² However, in contrast to our findings, several studies have found that small bowel wall thickness is not helpful in predicting the need for surgical intervention as the wall may become thin when gangrene develops.^{18–20} Use of the multiple CT features in combination is highly suggestive of the need for early surgical intervention.²³ This illustrates that a single finding on CT scan is unreliable and a cumulative score may be more appropriate.

Bowel ischemia causes an extensive inflammatory reaction resulting in erosions, free fluid, hemorrhage, ulcerations, and edema.²³ The constellation of findings associated with severe bowel obstruction produces multiple radiologic findings as reflected by the BIS. In this study, 66.3% of patients with BIS of 1 were successfully managed nonoperatively despite potential evidence of ischemia on CT. This illustrates

that although CT scan is valuable, these patients benefit from emphasis of multiple factors when making operative decisions. Only 16.7% of patients with BIS of 3 were successfully managed nonoperatively. Early operative intervention should be strongly considered in the patient with BIS of 3 or greater.

There were low numbers of patients with BIS of 3 or greater in this study. We suspect that most patients with BIS ≥ 4 were excluded from our study based on clinical indication for immediate operative intervention, such as peritonitis or hemodynamic instability. Therefore, BIS is most useful in patients with benign clinical presentation and 2 or 3 findings concerning for ischemia on CT scan. Bowel Ischemia Score may represent a helpful addition to current guidelines, which incorporate “signs of ischemia or strangulation” on CT, but do not specify the impact of specific findings or their cumulative effect.^{5,11} The finding of closed-loop obstruction should also be incorporated into the surgical decision-making process as it was independently associated with the need for surgical intervention. Similar helpful radiographic models exist in other disease processes. For example, RibScore predicts adverse outcomes and helps standardize assessment of rib fractures in the severely injured trauma patient.²⁴

There are several limitations to this study. It is a single-center study in an academic center with small sample size. We purposefully excluded patients without prior abdominal surgery or suspected neoplasm as the cause of obstruction and did not capture patients who may have been admitted to a medical service without EGS consultation. Small bowel resection was used as a surrogate for confirmed bowel ischemia, and there was a correlation between BIS and increasing rates of small bowel resection. Adding water-soluble contrast protocol during the study period may have affected the number of patients that went on to late (greater than 6 hours) operative intervention. However, the contrast is not administered until 12 hours after nasogastric tube placement, so early operative intervention is not likely to be affected. In addition, each CT scan was reviewed by a single attending radiologist which may introduce bias. However, a previous study showed that the differences in diagnostic performance of 3 radiologists were only 10% to 15% and not an issue for detecting strangulation.²⁴ Our model requires prospective validation among a national sample of patients with SBO.

In conclusion, we have developed a novel score, BIS, that accurately predicts the need for early operative intervention in patients with SBO without clinical indications for emergent surgery. Bowel Ischemia Score encompasses the cumulative signs of bowel ischemia on CT scan and more accurately predicts the need for early operation than the presence or absence of any one CT finding.

Supplementary Material

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

This study was presented as a poster presentation at the 78th Annual Meeting of AAST and Clinical Congress of Acute Care Surgery, September 18th, 2018 in San Diego, California.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Supplemental Material

Supplemental material for this article is available online.

References

Area under the receiver operating characteristic curve graphs for all patients. (A) Early surgery. (B) Small bowel resection. (C) Any surgery.

Characteristics of Historical vs. Prospective Patient Cohorts.

Abbreviations: SBO, small bowel obstruction; WBC, white blood cells.

Early surgical intervention defined as within 6 hours.

Elements of BIS and Predictive Value for Early Surgery.a

Abbreviation: BIS, Bowel Ischemia Score.

Early surgical intervention defined as within 6 hours.

Development of BIS Using Logistic Regression for Early Surgery.a

Abbreviations: BIS, Bowel Ischemia Score; SBO, small bowel obstruction; WBC, white blood cells.

Early surgical intervention defined as within 6 hours.

Outcomes by BIS.

Abbreviation: BIS, Bowel Ischemia Score.

Early Operative intervention defined as within 6 hours.

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